Lecture 9 – Dynamics of rotational motion

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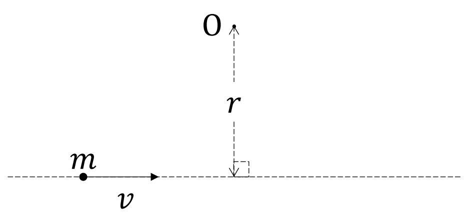
In the previous lecture we talked about the moment of inertia, which is the reluctance to start rotation. In this lecture we will discuss the dynamic of the rotation. We will learn how the motion of a **rigid body** rotating about a fixed axis changes in the course of time.

A few slides in this chapter come from Chapter-10 of Young's textbook.

1

Angular momentum of and torque on a particle

Consider a particle of mass m moving with speed v along a straight line with **perpendicular distance** r from point O:



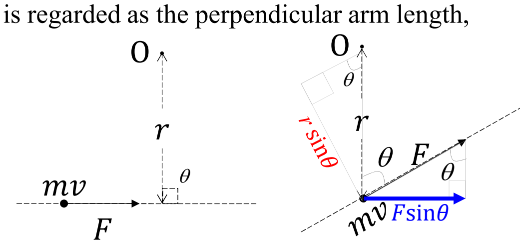
The **angular momentum** L of the particle about O is defined by $I\omega$. (Linear momentum is mv.)

$$L = I\omega \; = (mr^2)\omega = (m\cancel{r^2})\left(\frac{v}{\cancel{r}}\right)$$

$$\Rightarrow \boxed{L = rmv.}$$

This is a **vector quantity**.

If a force F acts on the particle as shown when r is regarded as the perpendicular arm length,



the **torque** τ on the particle about O is defined by $\boxed{\tau = F \times r \times \sin\theta.}$

If the force and arm length are not perpendicular to each other, we have to use the component of the F, i.e., $F \times \sin\theta$ as the applied force for the formulation of torque.

Torque is also a **vector quantity** given as

$$\tau = (F\sin\theta) \times r = F \times (r \times \sin\theta).$$

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Angular momentum of and torque on a particle (cont.)

By **Newton's equation of motion**,

$$F = ma = m\left(\frac{dv}{dt}\right) = \frac{d}{dt}(mv).$$

and **torque**

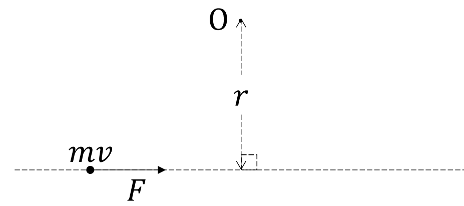
$$\tau = F \times r = \left(m\left(\frac{dv}{dt}\right)\right) \times r = \left(m\frac{r(d\omega)}{dt}\right) \times r$$

$$= \frac{(mr^2)\times(d\omega)}{dt} = \frac{(I\times(d\omega))}{dt} = \frac{dL}{dt}.$$

Here the application of torque will cause a change in **angular momentum**. **Torque** τ is equal to the **time rate change of angular momentum L of the particle**.

Force F is equal to the time rate change of linear momentum P of the particle.

If a force F acts on the particle and the distance to a reference point as shown:



Application of force will cause linear acceleration on the particle.

Application of torque will cause angular acceleration on the particle.

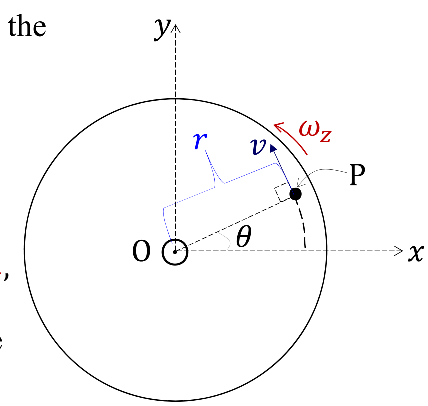
Moment of inertia and angular momentum of a rigid body rotating about a fixed axis

- For one small particle P of mass m shown on the figure the **angular momentum** is $(mr^2)\omega_z$.
- For the whole **rigid body**, the **angular momentum** of the rotation about a fixed axis with an **angular velocity** ω_z is

$$L_z = \sum(mr^2)\omega_z = \left(\sum mr^2\right)\omega_z = I\omega_z,$$

where the **moment of inertia** of the whole rigid body about the **axis of rotation** is

$$I = \sum mr^2 \quad \text{or} \quad I = \int r^2 dm.$$



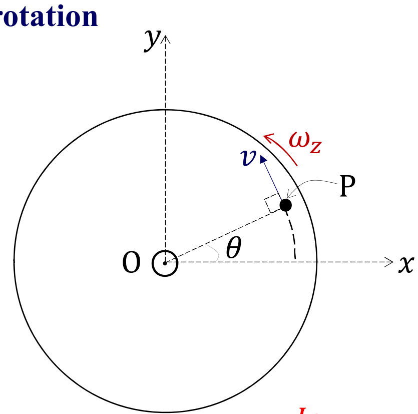
Angular momentum and torque on a rigid body rotating about a fixed axis

The **torque** on the rigid body about the **axis of rotation** is the change in angular momentum in 1 second

$$\tau_z = \frac{dL_z}{dt} = \frac{d(I\omega_z)}{dt} = I\left(\frac{d\omega_z}{dt}\right) = I\alpha_z$$

where α_z is the **angular acceleration** of the rigid body about the **axis of rotation**.

Torque τ is equal to the **time rate of change of angular momentum** L_z **of the whole rigid body**. It is the quantitative measure of the tendency of a **force** to cause or change a **rigid body's rotational motion**. So whenever there is torque, there will be angular acceleration (or deceleration).



$\tau = I\alpha$

$F = ma$

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Rotational kinetic energy of and torque on a rigid body rotating about a fixed axis

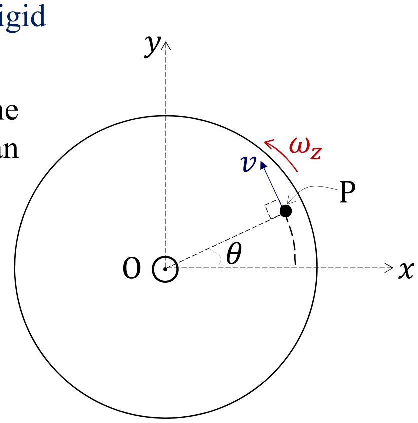
Recall the **rotational kinetic energy** of the rigid body rotating about a fixed axis with an angular velocity ω_z,

$$K = \frac{1}{2}I\omega^2 = \frac{1}{2}I\omega_z^2.$$

If a torque τ_z acts on the rigid body, ω or ω_z changes with time t:

$$\frac{d\omega_z}{dt} = \alpha_z$$

with $\tau_z = I\left(\dfrac{d\omega_z}{dt}\right) = I\alpha_z.$



net torque = moment of inertia x angular acceleration

net force = mass x acceleration

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Rotational kinetic energy of and torque on a rigid body rotating about a fixed axis (cont.)

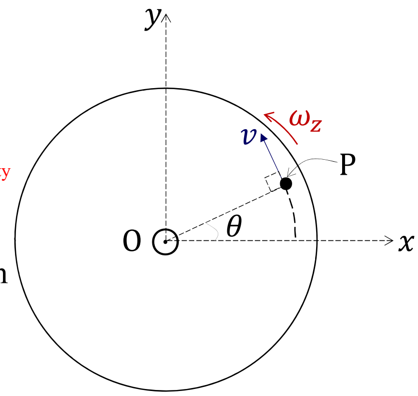
By **work-energy theorem**, the work done by the torque will be exhibited in the change of rotational kinetic energy. So the time rate of work done by the torque τ_z, i.e., the power

$$P = \frac{dK}{dt} = \frac{d\left(\frac{1}{2}I\omega^2\right)}{dt} = \frac{I}{2}\left(\frac{d(\omega^2)}{dt}\right) = \frac{I}{2}\left(\frac{d(\omega_z^2)}{dt}\right)$$

$$= \frac{I}{\cancel{2}}\left(\cancel{2}\omega_z\frac{d\omega_z}{dt}\right) = \left(I\frac{d\omega_z}{dt}\right)\omega_z = (I\alpha_z)\omega_z = \tau_z\omega_z.$$

Power = Torque x Angular Velocity

Power = Force x Linear Velocity



For times $t_1 < t_2$, the changes in rotation kinetic energy from K_1 to K_2 is contributed by the work done by the torque.

$$\int_{K_1}^{K_2} dK = \int_{t_1}^{t_2}\frac{dK}{dt}dt = \int_{t_1}^{t_2}(\tau_z\omega_z)dt$$

$$= \int_{t_1}^{t_2}\tau_z\left(\frac{d\theta}{\cancel{dt}}\right)\cancel{dt} = \int_{\theta_1}^{\theta_2}\tau_z d\theta$$

(Energy)

work done by torque = net torque x angular displacement along the same direction

work done by force = net force x displacement along the same direction

(Energy)

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Rotational kinetic energy of and torque on a rigid body rotating about a fixed axis (cont.)

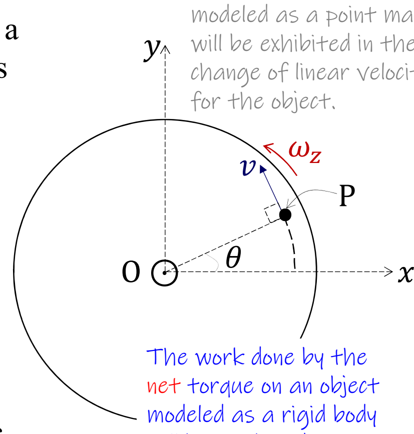
The work W done by the torque τ_z during a change in angular position from θ_1 to θ_2 is

$$W = \int_{\theta_1}^{\theta_2}\tau_z d\theta = K_2 - K_1 = \Delta K.$$

This is the parallel version of the **work-energy theorem**. Here, K_2 is the rotational kinetic energy of the rigid body at time t_2, and K_1 is that at time t_1.

In words, when torque does work on a rotating rigid body, the (rotational) kinetic energy changes by an amount equal to the work done.

The work done by the net force on an object modeled as a point mass will be exhibited in the change of linear velocity for the object.



The work done by the net torque on an object modeled as a rigid body will be exhibited in the change of angular velocity for the object.

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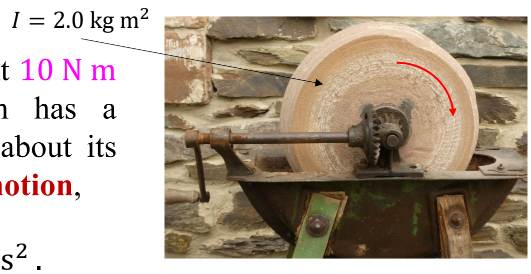
Example 1.

An electric motor exerts a constant 10 N m torque on a grindstone, which has a moment of inertia of 2.0 kg m^2 about its shaft. By **Newton's equation of motion**,

$$\tau_z = I\alpha_z \quad (F = ma)$$

$$\Rightarrow\ 10 = 2\alpha_z \Rightarrow \alpha_z = 5\ \text{rad/s}^2.$$



The angular acceleration of the grindstone about its shaft is $\alpha_z = 5\ \text{rad/s}^2$.

If the grindstone starts from rest, in $t = 8.0$ s its angular velocity about its shaft is

$$\omega_z = \omega_0 + \alpha_z t = 0 + 5 \times 8.0 = 40\ \text{rad/s, and}$$

the rotational kinetic energy of the grindstone is $KE_{\text{rotation}} = \dfrac{1}{2} I\omega_z^2 = \dfrac{1}{2} \times 2.0 \times 40^2 = 1600$ J.

By the **work-energy theorem**, the work W done by the motor in 8.0 s is exhibited by the rotation of the grindstone. So $W = KE - 0 = 1600$ J.

The average power delivered by the motor is $P_{\text{average}} = \dfrac{W}{t} = \dfrac{1600}{8} = 200$ W.

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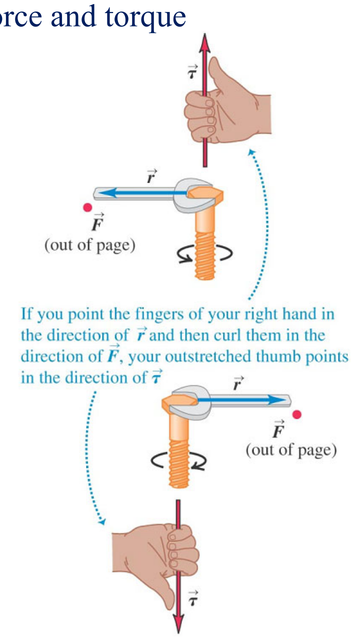
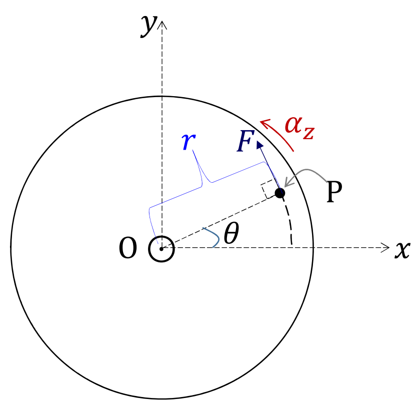
Force and torque





The **torque** of the **force** F exerted on the rigid body about the **axis of rotation** is $\tau_z = F \times r$, where r is the perpendicular arm length to the force F. The SI unit of torque is the newton-metre, N m.

This N m is not energy. Why? **Torque** is a vector!!

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Force and torque (cont.)

Remarks

- Recall that when a **rigid body** of mass M, moving on the xy-plane, is acted on by (**external**) **forces**,

$$F_x = Ma_{cm,x}, \quad F_y = Ma_{cm,y}$$

with (F_x, F_y) the (**external**) **net force**, and $(a_{cm,x}, a_{cm,y})$ **acceleration of center-of-mass**, it does not matter where the forces act when we add them to obtain the net force as the objects are modeled as a point mass.

- What happen if the (**external**) **net force** is **zero**? Newton's 1st Law (Law of Inertia!!)
  This will result in (**external**) **net torque = 0.** What will be the outcome?

- When a **rigid body** with moment of inertia I, rotating about the z-axis, is acted on by (**external**) **torques**, it obeys the Newton's 2nd Law $\tau_z = I\alpha_z$ ($F = ma$!!)
  where τ_z is the (**external**) **net torque**, and α_z the **angular acceleration** about the z-axis. It is now important to note where the (**external**) **forces** act when we add their contributions to the net torque. The position will affect the arm length, thus affect the torque.

Force and torque (cont.)

In the absence of net external force linear momentum is conserved.

In the absence of net external torque, angular momentum is conserved.

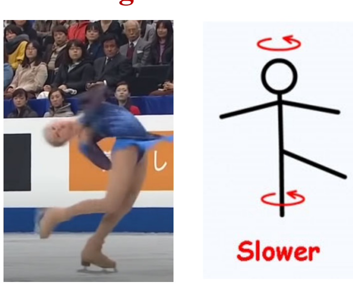
Remarks (cont.).

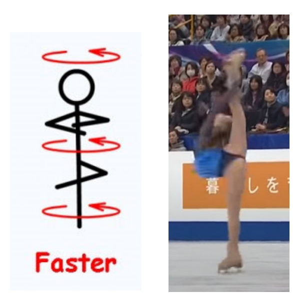
- If the (**external**) **net torque** is **zero**,

$$I\alpha_z = I\left(\frac{d\omega_z}{dt}\right) = \frac{d}{dt}(I\omega_z) = F \times r = 0 \quad \Rightarrow \quad I\omega_z = \text{constant.}$$

- In general, when the (**external**) **net torque** acting on a system is **zero**, the **total angular momentum of the system is constant** (**conserved**). This is the **conservation of angular momentum**.

$I_i\omega_i$

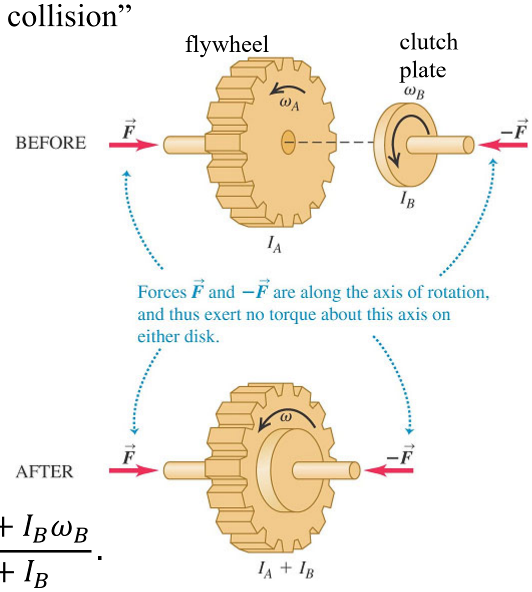




$I_f\omega_f$

Example 2. A rotational "completely inelastic collision"

Consider two disks: an engine flywheel A and a clutch plate B attached to a transmission shaft. Their moments of inertia are I_A and I_B. Initially, they are rotating with constant angular speeds ω_A and ω_B, respectively. We push the disks together with forces acting along the axis, so as not to apply any torque on either disk. The disks rub against each other and eventually reach a common angular speed ω.



By **conservation of angular momentum** before and after the contact,

$$I_A\omega_A + I_B\omega_B = (I_A + I_B)\omega \;\Rightarrow\; \omega = \frac{I_A\omega_A + I_B\omega_B}{I_A + I_B}.$$

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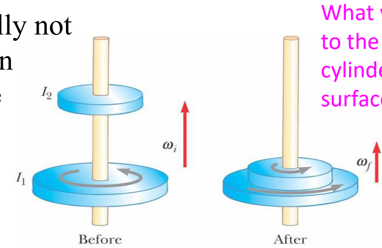
Question. A cylinder with moment of inertia I_1 rotates about a vertical, frictionless axle with angular velocity ω_i. A second cylinder, this one having moment of inertia I_2 and initially not rotating, drops onto the first cylinder. Because of friction between the surfaces, the two eventually reach the same angular velocity ω_f.



What will happen to the smaller cylinder if the surface is smooth?

(i) Calculate ω_f.

Answer:

From conservation of angular momentum for the system of two cylinders:

$$I_1\omega_i = (I_1 + I_2)\omega_f \;\Rightarrow\; \omega_f = \boxed{\frac{I_1}{I_1 + I_2}\omega_i}$$

(ii) Show that the kinetic energy of the system decreases in this interaction and calculate the ratio of the final to the initial rotational energy.

Answer: $K_i = \frac{1}{2}I_1\omega_i^2$

$$K_f = \frac{1}{2}(I_1 + I_2)\omega_f^2$$

$$\Rightarrow \frac{K_f}{K_i} = \frac{\cancel{\tfrac{1}{2}}\cancel{(I_1 + I_2)}\left(\frac{I_1}{I_1 + I_2}\cancel{\omega_i}\right)^{\cancel{2}}}{\cancel{\tfrac{1}{2}}I_1\omega_i^2}$$

Why loss in rotational KE ?

$$= \boxed{\frac{I_1}{I_1 + I_2} \text{ which is less than 1}}$$

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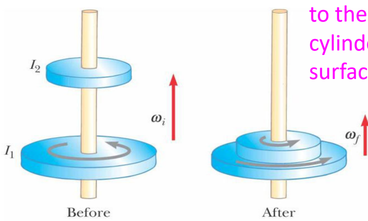

What will happen to the smaller cylinder if the surface is smooth?

Answer:

Rotation of the smaller cylinder will not start if there is no friction on the surface.

But the inference is only one way, that is, if there is a rotation, it does not necessarily mean that the surface is rough.

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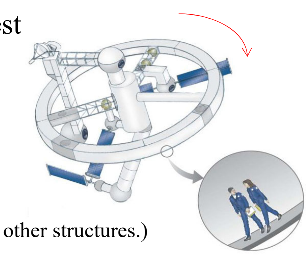
Question. A rotating-wheel space station, or called a Donut City, is constructed in the shape of a hollow ring of mass 5.00×10^4 kg. Members of the crew walk on a deck formed by the inner surface of the outer cylindrical wall of the ring, with radius 100 m. At rest when constructed, the ring is rotated about its axis so that the people inside experience an effective free-fall acceleration equal to g. The rotation is achieved by firing two small rockets attached tangentially to opposite points on the outside of the ring. (Assumption: The moment of inertia for the Donut City will only focus on the ring due to the huge mass. You can ignore the mass of the 3 supporting pillars and other structures.)



(i) What angular momentum does the space station acquire?

Answer:

The centripetal acceleration is $a_c = g = \dfrac{v^2}{r} = \dfrac{(r\omega)^2}{r} = r\omega^2$

$$\Rightarrow \omega = \sqrt{\frac{g}{r}} = \sqrt{\frac{(9.80\ \text{m/s}^2)}{100\ \text{m}}} = 0.313\ \text{rad/s}$$

$$I = \sum m_i r_i^2 = \left(\sum m_i\right) r^2 = Mr^2$$
$$= 5\times10^4\ \text{kg}(100\ \text{m})^2 = 5\times10^8\ \text{kg}\cdot\text{m}^2$$

Angular Momentum :

$$L = I\omega$$
$$= 5\times10^8\ \text{kg}\cdot\text{m}^2 \times 0.313\ \text{rad/s}$$
$$= \boxed{1.57\times10^8\ \text{kg}\cdot\text{m}^2/\text{s}}$$

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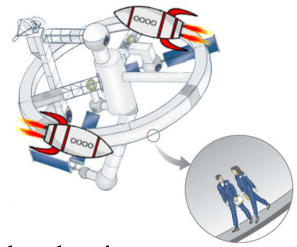
(ii) How long must the rockets be fired if each exerts a thrust of 125 N?

Answer:

Force x Time = Changes in Linear Momentum

Torque x Time = Changes in Angular Momentum



$$\Rightarrow \Delta t = \frac{L_f - 0}{\sum F_i \times r_i} = \frac{1.57\times10^{8}\ \text{kg}\cdot\text{m}^2/\text{s}}{(125\ \text{N}\times100\ \text{m})+(125\ \text{N}\times100\ \text{m})}$$

$$= \boxed{6.26\times10^{3}\ \text{s}} = 1.74\ \text{h}$$

(iii) Prove that the total torque on the ring multiplied by the time interval found in part (ii), is equal to the change in angular momentum found in part (i).

Answer:

$$\sum \tau = I\alpha = \frac{I\omega_f - I\omega_i}{\Delta t}$$

$$\Rightarrow \sum \tau\ \Delta t = I\omega_f - I\omega_i = L_f - L_i = \Delta(I\omega)$$

This is the angular impulse-angular momentum theorem.

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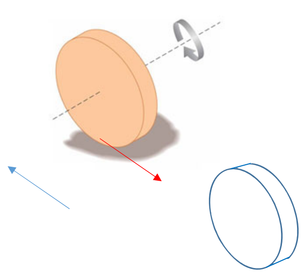
Question. A uniform solid disk of radius *r* and mass *m* is set into rotation with an angular speed ω_i about an axis through its center. While still rotating at this angular speed, the disk is placed into contact with a horizontal rough surface where the coefficient of kinetic friction is μ_k, and is immediately released with no linear velocity at the center of the disk as shown in the following figure.

Calculate the linear displacement of the solid disk from the first contact point to the position as soon as pure rolling occurs.

Answer:

What is the cause for the linear displacement of the disk?



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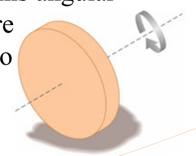
Translation (Velocity v) is greater than Rotation ($r\omega$)

Rotation ($r\omega$) is greater than Translation (Velocity v)

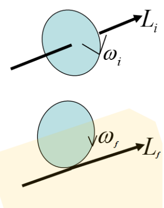
Question. A uniform solid disk of radius r and mass m is set into rotation with an angular speed ω_i about an axis through its center. While still rotating at this angular speed, the disk is placed into contact with a horizontal rough surface where the coefficient of kinetic friction is μ_k, and is immediately released with no linear velocity at the center of the disk as shown in the following figure.



Calculate the linear displacement of the solid disk from the first contact point to the position as soon as pure rolling occurs.

Answer:

In the absence of net external torque, angular momentum is conserved.



$$I\omega_i = \left(I + mr^2\right)\omega_f$$

$$\Rightarrow \omega_f = \frac{I}{I + mr^2}\omega_i = \left(\frac{\frac{1}{2}mr^2}{\frac{1}{2}mr^2 + mr^2}\right)\omega_i = \frac{\omega_i}{3}$$

So $v_f = r\omega_f = r\left(\frac{\omega_i}{3}\right)$

Let f be the friction on the surface. From the linear impulse:

$$f \times \Delta t = mv_f - mv_i$$

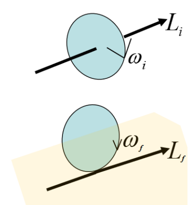
$$\Rightarrow \Delta t = \frac{mv_f - \cancel{mv_i}^{\,0}}{f} = \frac{mv_f}{f}$$

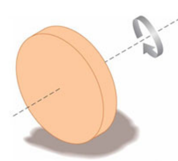
$$= \frac{\cancel{m}\left(r\left(\frac{\omega_i}{3}\right)\right)}{\mu_k \cancel{m} g} = \frac{r\omega_i}{3\mu_k g}$$

From $\Delta x = \left(\frac{v_f + v_i}{2}\right)t = \left(\frac{\frac{r\omega_i}{3} + 0}{2}\right) \times \frac{r\omega_i}{3\mu_k g}$

$$= \frac{r^2\omega_i^2}{18\mu_k g}$$

In the absence of net external torque, angular momentum is conserved.





$$I\omega_i = \left(I + mr^2\right)\omega_f$$

$$\Rightarrow \omega_f = \frac{I}{I + mr^2}\omega_i = \left(\frac{\frac{1}{2}mr^2}{\frac{1}{2}mr^2 + mr^2}\right)\omega_i = \frac{\omega_i}{3}$$

Changes in rotational kinetic energy:

$$\underbrace{I}_{\text{disk}} = \frac{mr^2}{2} \Rightarrow mr^2 = 2I$$

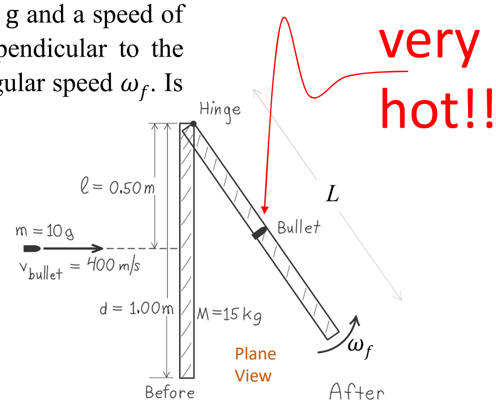
$$\frac{1}{2}\left(\frac{mr^2}{2} + mr^2\right)\omega_f^2 - \frac{1}{2}\left(\frac{mr^2}{2}\right)\omega_i^2 = \frac{1}{2}(I + 2I)\omega_f^2 - \frac{1}{2}I\omega_i^2$$

$$= \frac{1}{2}I\left(3\omega_f^2 - \omega_i^2\right) = \frac{1}{2}I\left(3\left(\frac{\omega_i}{3}\right)^2 - \omega_i^2\right)$$

$$= \frac{1}{2}I\omega_i^2\left(\frac{1}{3} - 1\right) < 0$$

Loss in rotational kinetic energy so the table becomes very hot!!

Problem. A door 1.00 m wide, of mass 15.0 kg, can rotate freely about a vertical axis through its hinges. A bullet with a mass of 10.0 g and a speed of 400 m/s strikes the centre of the door, in a direction perpendicular to the plane of the door, and embeds itself there. Find the door's angular speed ω_f. Is kinetic energy conserved? Prove it quantitatively.



Answer:

Initial angular momentum in counter-clockwise direction:

$$I\omega_i = \left(m\ell^2\right)\left(\frac{v_{\text{bullet}}}{\cancel{\ell}}\right) = m\ell v_{\text{bullet}}$$

$$v_{\text{bullet}} = r\omega_i \Rightarrow \omega_i = \frac{v_{\text{bullet}}}{r}$$

By conservation of angular momentum, $I\omega_i = I_{\text{total}}\omega_f$.

$$\Rightarrow m\ell v_{\text{bullet}} = \left(m\ell^2 + \frac{1}{3}ML^2\right)\omega_f = \left(m\ell^2 + \frac{1}{3}M\left(2\ell\right)^2\right)\omega_f$$

$$\Rightarrow 0.01\times 0.5\times 400 = \left(0.01\times 0.5^2 + \frac{1}{3}\times 15\times (2\times 0.5)^2\right)\omega_f$$

$$\Rightarrow \omega_f = 0.400 \text{ rad/s}$$

Kinetic energy not conserved!!

$$\text{Initial translational KE} = \frac{1}{2}mv_{\text{bullet}}^2 = \frac{1}{2}\times 0.01\times 400^2 = 800\text{ J}$$

$$\text{Final rotational KE} = \frac{1}{2}I_{\text{total}}\omega_f^2 = \frac{1}{2}\left(m\ell^2 + \frac{1}{3}ML^2\right)\omega_f^2 = \frac{1}{2}\left(0.01\times 0.5^2 + \frac{1}{3}\times 15\times 1^2\right)0.4^2 = 0.400\text{ J}$$

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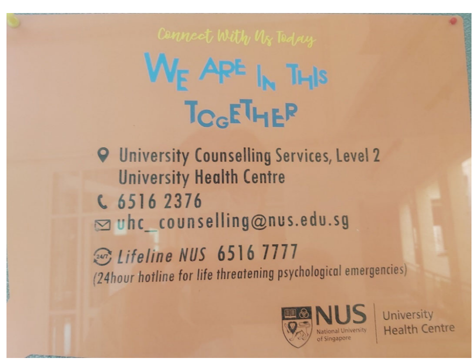
Helplines to de-stress

We are not superman/superlady. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019

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